Exact tridiagonal interval determinants and solution hulls are NP-hard

Repository IV-02 and IV-04. Claimed resolutions; independent review pending.

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Independently reviewed resolution. This is a complexity classification; it does not establish P unequal to NP. The dated independent agent review supersedes the original pending-review header. Agent review is not external human peer review or formal certification. The source archive describes these as AI-generated drafts; authorship is attributed at the submitter's request.

Abstract

A reduction from PARTITION encodes signed sums of small rational rotation angles in the continuant of a tridiagonal interval matrix. It proves NP-completeness of the upper determinant threshold problem, and NP-hardness of exact determinant-range and exact solution-hull computation. The reduction uses independent interval entries, fixed superdiagonal entries equal to 1, and an everywhere nonsingular interval family. For the hull result the right-hand side is the single point $-e_n$. General exact-output upper bounds are also recorded, so each requested polynomial-time existence question is equivalent to $\mathsf{P} = \mathsf{NP}$.

1 Scope

The exact determinant-range and solution-hull problems for rational tridiagonal interval matrices are the subjects of IV-02 and IV-04 [1, 2]. Polynomial algorithms for important special subclasses do not resolve the general questions [3, 4, 5]. The reduction below does not assert strong NP-hardness: the rational gap can be small in the numerical value of the input.

2 Rational rotations as continuants

For rational a, β, put

$$M(a,\beta) = \begin{pmatrix} a & -\beta \\ 1 & 0 \end{pmatrix}.$$

Given a PARTITION instance with positive integer weights $w_1, \dots, w_m$, let

$$W = \sum_i w_i, \quad \delta = \frac{1}{10W^2}, \quad t_i = \delta w_i, \quad c_i = \frac{1-t_i^2}{1+t_i^2}, \quad s_i = \frac{2t_i}{1+t_i^2}.$$

Here $c_i > 0$, $c_i^2 + s_i^2 = 1$, and

$$R_i = \begin{pmatrix} c_i & -s_i \\ s_i & c_i \end{pmatrix} = M(-s_i/c_i, -1/c_i) M(s_i, -c_i)$$

is a rotation through $\theta_i = 2\arctan(t_i)$. Also,

$$M(0,-1)M(0,q) = \operatorname{diag}(1,-q), \qquad q \in [-1,1].$$

Thus a pair of companion matrices implements a rotation, and another pair implements an independent reflection-or-identity gate at its endpoints.

Construct $N = 4m-2$ layers, in order of application, as follows:

j	a_j	β_j	range
$4i-3$	s_i	$-c_i$	$i = 1,\dots,m$
$4i-2$	$-s_i/c_i$	$-1/c_i$	$i = 1,\dots,m$
$4i-1$	0	q_i	$i = 1,\dots,m-1, \quad q_i \in [-1,1]$
$4i$	0	-1	$i = 1,\dots,m-1.$

Let $T(q)$ have diagonal a_j, all superdiagonal entries equal to 1, and subdiagonal entry $T_{j,j-1} = \beta_j$ for $j \ge 2$. The unused β_1 is only a notational convenience in the transfer identity. The $m-1$ intervals $q_i \in [-1,1]$ are distinct matrix entries and are mutually independent.

The continuant recurrence for leading determinants is

$$p_{-1} = 0, \quad p_0 = 1, \quad p_j = a_j p_{j-1} - \beta_j p_{j-2}.$$

Therefore $\det T = p_N$ is the first coordinate of

$$M(a_N,\beta_N)\cdots M(a_1,\beta_1)(1,0)^T.$$

At a gate vertex, put $\epsilon_i = -q_i \in \{-1,1\}$. The successive vector angles satisfy

$$\phi_1 = \theta_1, \qquad \phi_{i+1} = \theta_{i+1} + \epsilon_i\phi_i.$$

Consequently the endpoint determinants are exactly

$$\cos\Big(\sum_{i=1}^{m}\sigma_i\theta_i\Big), \qquad \sigma_i \in \{-1,1\}, \quad \sigma_m = 1. \tag{1}$$

Every such pattern occurs: take $\epsilon_i = \sigma_i\sigma_{i+1}$. Fixing the last sign loses no zero signed sum, since a simultaneous sign reversal preserves zero.

3 Separation and regularity

The elementary integral bound

$$0 \le 2t - 2\arctan t \le \frac{2}{3}t^3 \qquad (t \ge 0)$$

gives

$$E := \sum_i |\theta_i - 2\delta w_i| \le \frac{2}{3}\delta^3 W^3 = \frac{\delta}{150W} \le \frac{\delta}{10}.$$

Also $\sum_i \theta_i \le 2\delta W = 1/(5W) \le 1/5$. Every determinant in (1) is therefore positive. A determinant is multiaffine in independent entries; throughout a box it is a convex combination of its vertex values. Thus every matrix in the entire interval family is nonsingular, not just its vertices. The determinant maximum is attained at a vertex.

If PARTITION has a solution, choose its signs with $\sigma_m = 1$. Then the angle in (1) has absolute value at most $\delta/10$, giving

$$\max_q \det T(q) \ge 1 - \frac{\delta^2}{200}.$$

If there is no partition, each integer signed weight sum has absolute value at least 1. Every endpoint angle then has absolute value at least $2\delta - E \geq 19\delta/10$, and at most $1/5$. For $|x| \leq 1$, Taylor's inequality gives $\cos x \leq 1 - x^2/3$. Hence

$$\max_q \det T(q) \leq 1 - \frac{361}{300}\delta^2 < 1 - \delta^2.$$

The rational threshold

$$\tau = 1 - \frac{\delta^2}{2}$$

strictly separates the two cases.

All numbers constructed above have $O(\log W)$ bits per entry. For example c_i and s_i use integers of sizes polynomial in W, not an exponentially long list of repeated rotations. There are $O(m)$ rows and interval entries. The reduction is polynomial in the binary input length.

Theorem 1. *Deciding whether the maximum determinant of a rational tridiagonal interval matrix is at least a rational threshold is NP-complete. NP-hardness holds even when every matrix in the interval family is nonsingular and every superdiagonal entry is fixed at 1. In particular exact determinant-range computation is NP-hard.*

Proof. Hardness is the reduction above from the standard NP-complete PARTITION problem [6]. Membership in NP follows from multiaffinity: a vertex is a polynomial-size certificate, and its rational determinant is computable exactly in polynomial time. □

4 Transferring hardness to a solution hull

For a tridiagonal $N \times N$ matrix with all superdiagonal entries 1, the corner cofactor identity is

$$(T^{-1})_{1N} = \frac{(-1)^{N+1}}{\det T}.$$

Indeed, deleting row N and column 1 leaves a triangular matrix with diagonal entries all 1. Our $N = 4m - 2$ is even and all determinants are positive. Take the point right-hand side $b = -e_N$. Then

$$x_1 = (T^{-1}b)_1 = \frac{1}{\det T} > 0.$$

The lower endpoint of coordinate 1 of the exact interval solution hull is therefore

$$\underline{x}_1 = \frac{1}{\max_q \det T(q)}.$$

It decides the determinant threshold by comparison with $1/\tau$.

Theorem 2. *Exact solution-hull computation for rational tridiagonal interval systems is NP-hard even for regular interval matrices, point right-hand side $-e_N$, and fixed superdiagonal entries all equal to 1. These instances have nonempty bounded solution sets.*

5 Exact-output upper bounds

For completeness, both general exact-output problems are in $\mathsf{FP}^{\mathsf{NP}}$. This also makes precise the "unless $\mathsf{P} = \mathsf{NP}$" qualification.

For determinants, multiply all vertex entries by a common denominator D whose bit length is polynomial in the input. Let H bound the absolute values of the resulting integers. The scaled determinant is an integer of absolute value at most $n!H^n$. Binary search with the NP vertex-threshold oracle finds its exact maximum and minimum in polynomially many queries; divide by D^n.

For hulls, write the interval data as $[C-R, C+R]$ and $[b_c-b_r, b_c+b_r]$. Rowwise independent interval evaluation gives the exact solution set

$$\Sigma = \{x : |Cx - b_c| \le R|x| + b_r\} = \bigcup_{s\in\{-1,1\}^n} P_s,$$

where

$$P_s = \{x : D_s x \ge 0,\ (C - RD_s)x \le b_c + b_r,\ (-C - RD_s)x \le -b_c + b_r\}.$$

Each P_s is a rational pointed polyhedron, since it is contained in an orthant. Nonempty such polyhedra have vertices, and finite coordinate optima are attained at vertices. After a common denominator is cleared in the coefficients and right-hand sides, let H bound their integer sizes and put $Q = n!H^n$, increasing H to at least 1. Cramer's rule bounds every vertex coordinate numerator in absolute value by Q and its denominator by Q.

Nonemptiness is in NP: guess s and a polynomial-bit feasible vertex. Unboundedness of coordinate i is in NP: guess a feasible sign polyhedron and a rational recession vector normalized to have ith coordinate 1 (or -1). Such certificates have polynomial encoding length by standard rational-polyhedron bounds. Because the union has finitely many members, an unbounded coordinate in the union is unbounded in at least one member.

When an endpoint is finite, the query $\exists x \in \Sigma : x_i \ge t$ is in NP by the same sign-polyhedron certificate argument, including the extra rational inequality. Binary search brackets the endpoint in an interval of width less than $1/(2Q^2)$ inside $[-Q, Q]$. At most one rational with denominator at most Q is in that bracket. Continued-fraction rational reconstruction recovers it in polynomial time. Lower endpoints are handled by $-x_i$. Empty sets and infinite endpoints are covered by the earlier oracle queries.

Thus a polynomial-time algorithm exists for either complete exact-output task if $\mathsf{P} = \mathsf{NP}$. The hardness theorems prove the converse. These upper bounds are included as standard complexity completion, not as an independent novelty claim.

References

[1] OpenProblemsInNLA, problem IV-02. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-02`.

[2] OpenProblemsInNLA, problem IV-04. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-04`.

[3] J. Horáček, M. Hladík and J. Matějka, *Determinants of interval matrices*, Electronic Journal of Linear Algebra 33 (2018), 99–112. `https://doi.org/10.13001/1081-3810.3719`.

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[6] R. M. Karp, *Reducibility among combinatorial problems*, in Complexity of Computer Computations (1972), 85–103. `https://doi.org/10.1007/978-1-4684-2001-2_9`.